PS Analysis in Data Science: Problem Set 7

Problem 1. Let $f : (-\frac{1}{2}, \infty) \times (-\frac{1}{2}, \infty) \to \mathbb{R}$ be defined by

$$f(x, y) = \log(1 + x + y).$$

(a) Determine the Taylor polynomial of order 3 of f at $(0, 0)$.

(b) Use your result to approximate $f(0.1, 0.1)$.

(c) Plot heatmaps of f, of the Taylor polynomial of order 3, and of the difference of the two in $[0, 1]^2$.

Problem 2. Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x, y) = x^4 + y^4 - 2x^2 - 2y^2.$$

(a) Compute the gradient $\nabla f(x, y)$.

(b) Determine all critical points of f and classify them as local minima, local maxima, or saddle points via the Hessian matrix.

(c) Implement gradient descent for f in Julia. Experiment with several starting values and describe which critical points the algorithm converges to.

Problem 3. Let V and W be a normed spaces and let $f_1, \dots, f_n : V \to \mathbb{R}$ be convex. Prove that the following functions are also convex.

(a) $\sum_{i=1}^{n} \alpha_i f_i$ for $\alpha_1, \dots, \alpha_n \geq 0$.

(b) $g : V \to \mathbb{R}$ given by $g(x) = \max_{i=1,\dots,n} f_i(x)$.

(c) If $A : W \to V$ is linear and $b \in V$, then $g : W \to \mathbb{R}$ given by

$$g(x) = f_1(Ax - b)$$

is also convex.

Problem 4. Let $n = 1000$, $d = 3$, and define

$$a_i = \begin{pmatrix} 1 \\ t_i \\ t_i^2 \end{pmatrix}, \qquad t_i = \frac{i}{n}, \qquad x_{\text{true}} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

Set $b_i = a_i^\top x_{\text{true}}$ and consider

$$f(x) = \frac{1}{n} \sum_{i=1}^{n} (a_i^\top x - b_i)^2.$$

(a) Show that

$$\nabla f(x) = \frac{2}{n} \sum_{i=1}^{n} (a_i^\top x - b_i) a_i.$$

(b) Implement the following three methods with

$$x_0 = 0, \qquad \eta = 0.01, \qquad N = 50000, \qquad m = 8.$$

$$x_{k+1}^{\text{GD}} = x_k - \eta \nabla f(x_k), \tag{1}$$

$$x_{k+1}^{\text{SGD}} = x_k - \eta\, 2(a_{i_k}^T x_k - y_{i_k}) a_{i_k}, \qquad i_k = 1 + (k \bmod n), \tag{2}$$

$$x_{k+1}^{\text{MB}} = x_k - \eta \frac{1}{m} \sum_{i \in B_k} 2(a_i^T x_k - y_i) a_i, \tag{3}$$

where

$$B_k = \{1 + (km + j \bmod n) : j = 0, \ldots, m-1\}.$$

(c) Plot $f(x_k)$ for all three methods.